

THE SPECTRAL MEASURE OF THE SAME-CYCLE INDICATOR UNDER RANDOM TRANSPOSITIONS

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ABSTRACT. Let $u \neq v$ be two distinguished points of $[N]$ and let $\varepsilon(\sigma) = 2 \cdot \mathbf{1}\{u \text{ and } v \text{ lie in a common cycle of } \sigma\} - 1$ on the symmetric group S_N . We show that the spectral measure of ε with respect to the random-transposition walk has an explicit closed form: it is supported on the partitions $\lambda = (a, b, 1^j)$ with $a \geq b \geq 1$, and there

$$\|\Pi_\lambda \varepsilon\|^2 = \frac{2(a-b+1)^2}{(a+j+1)^2(b+j)^2} (1 - r_\lambda),$$

where r_λ is the normalised character of a transposition. The proof is a few lines: $\mathbf{1}\{u \sim_\sigma v\}$ is half the increment of the number of cycles under right multiplication by (uv) , and the Fourier coefficients of the cycle-count function are read off the content polynomial. Consequences include the exact two-point correlation $\langle \varepsilon, P^m \varepsilon \rangle$ of the same-cycle status along the walk, the bound $|\langle \varepsilon, P^m \varepsilon \rangle| \leq 12/m + O((\log m/m + 1/N)^4)$ with absolute constants, the limiting spectral density $4(\alpha - \beta)^2 / ((1 - \alpha)(1 - \beta)) d\alpha d\beta$ (so that the correlation at lag m is $2/(m+1) - 3/(2m^2) + O(m^{-3})$ as $N \rightarrow \infty$), and the probability $(N+1)(N+2)/(6N(N-1))$ that one random transposition changes the same-cycle status of two points of a uniformly random permutation. The same spectral measure governs every conjugation-invariant random walk.

1. INTRODUCTION AND MAIN RESULT

Let $N \geq 2$ and let $\sigma_0, \sigma_1, \sigma_2, \dots$ be the random-transposition walk on S_N : $\sigma_{t+1} = \sigma_t \tau_{t+1}$ with $\tau_1, \tau_2, \dots$ independent uniform transpositions. Fix two points $u \neq v$ of $[N]$ and write $u \sim_\sigma v$ if u and v lie in a common cycle of σ . The evolution of the pair (u, v) under the walk is the simplest instance of the coagulation–fragmentation dynamics of the cycle structure studied by Schramm [10], Berestycki–Durrett [3], Berestycki–Schramm–Zeitouni [5] and many others: a transposition whose two points lie in one cycle splits it, and merges two cycles otherwise, and the status of (u, v) changes exactly when the transposition separates them or joins their cycles. Under the uniform measure $\mathbb{P}[u \sim v] = \frac{1}{2}$, so the natural two-point quantity is the autocorrelation of the sign

$$\varepsilon(\sigma) = 2 \cdot \mathbf{1}\{u \sim_\sigma v\} - 1 \in \{\pm 1\}$$

along the stationary walk, $\mathbb{E}[\varepsilon(\sigma_0)\varepsilon(\sigma_m)]$ with σ_0 uniform.

The random-transposition walk is driven by a central element of the group algebra, so it is diagonalised by the irreducible characters of S_N , with eigenvalue on the λ -isotypic component

$$(1) \quad r_\lambda = \frac{\chi_\lambda(\tau)}{d_\lambda} = \frac{2}{N(N-1)} \sum_{\square \in \lambda} \text{cont}(\square),$$

where $\text{cont}(i, j) = j - i$ is the content of the box in row i and column j [7]. Consequently every autocorrelation of a function f on S_N is a moment of its spectral measure $\nu_f(\lambda) = \|\Pi_\lambda f\|^2$:

$$\langle f, P^m f \rangle = \sum_{\lambda \vdash N} r_\lambda^m \nu_f(\lambda), \quad \|P^d f\|^2 = \sum_{\lambda \vdash N} r_\lambda^{2d} \nu_f(\lambda).$$

Here $L^2(S_N)$ carries the uniform measure, $(Pf)(\sigma) = \binom{N}{2}^{-1} \sum_{\tau} f(\sigma\tau)$, and $\Pi_\lambda = \frac{d_\lambda}{N!} \sum_{\rho} \chi_\lambda(\rho) R(\rho)$ with $(R(\rho)f)(\sigma) = f(\sigma\rho)$ is the projection onto the λ -isotypic component of the right-regular representation. For class functions the spectral measure is simply the square of the Fourier coefficient, and for the cycle-count statistics it is known: Alon and Kozma [1, Thm. 3] expand the number of k -cycles over the shapes $(N-k, k-i, 1^i)$, and Berestycki and Kozma [4] use this expansion to study the cycle structure of the interchange process (see also [2]). The function ε is not a class function. Nevertheless its spectral measure is explicit, because $\mathbf{1}\{u \sim v\}$ is a first difference of the cycle count.

Theorem 1.1. $\nu_\varepsilon(\lambda) \neq 0$ only for $\lambda = (a, b, 1^j)$ with $a \geq b \geq 1$ and $a + b + j = N$, and for these shapes

$$(2) \quad \nu_\varepsilon(\lambda) = \|\Pi_\lambda \varepsilon\|^2 = 2c_\lambda^2(1-r_\lambda), \quad c_\lambda = \frac{a-b+1}{(a+j+1)(b+j)}, \quad r_\lambda = \frac{a(a-1)+b(b-3)-j(j+3)}{N(N-1)}.$$

Here $c_\lambda = |\langle c, \chi_\lambda \rangle|$ is the modulus of the Fourier coefficient of the number-of-cycles function $c(\sigma)$.

Thus, for σ_0 uniform,

$$(3) \quad \mathbb{E}[\varepsilon(\sigma_0)\varepsilon(\sigma_m)] = \sum_{\substack{a \geq b \geq 1 \\ a+b+j=N}} \frac{2(a-b+1)^2}{(a+j+1)^2(b+j)^2} (1-r_\lambda) r_\lambda^m,$$

and $\sum_\lambda \nu_\varepsilon(\lambda) = \|\varepsilon\|^2 = 1$. Table 1 gives some values. Section 2 proves the theorem, Section 3 draws the consequences announced in the abstract, and Section 4 records the generalisations (other conjugation-invariant walks, non-stationary starts) that the proof gives for free. The computation was motivated by a problem on random Latin squares, where the status of two marked rows in the cycle structure of a column-pair permutation is randomised by switching a pool of intercalates [8]; we say a word about this in Remark 4.1.

m	1	2	4	8	16	64	256	1024
$N = 4000$	0.6663	0.5107	0.3418	0.2037	0.1121	0.02992	0.00727	0.00149
$N = \infty$	2/3	23/45	0.3422	0.2042	0.1125	0.03042	0.00776	0.00195
$m \cdot (N = \infty)$	0.667	1.022	1.369	1.634	1.801	1.947	1.986	1.997

TABLE 1. The autocorrelation $\mathbb{E}[\varepsilon(\sigma_0)\varepsilon(\sigma_m)]$ of the same-cycle sign along the stationary random-transposition walk, from (3) and from the limit (7).

2. PROOF OF THEOREM 1.1

Step 1: the indicator is a difference of the cycle count. Let $c(\sigma)$ be the number of cycles of σ . Right multiplication by the transposition (uv) sends $u \mapsto \sigma(v)$ and $v \mapsto \sigma(u)$ and fixes every other image. If u and v lie on one cycle $u \rightarrow \sigma(u) \rightarrow \cdots \rightarrow v \rightarrow \sigma(v) \rightarrow \cdots \rightarrow u$, the new permutation closes the two cycles $u \rightarrow \sigma(v) \rightarrow \cdots \rightarrow u$ and $v \rightarrow \sigma(u) \rightarrow \cdots \rightarrow v$; if they lie on different cycles, these are spliced into one. Hence $c(\sigma(uv)) - c(\sigma) = +1$ if $u \sim_\sigma v$ and -1 otherwise, i.e.

$$(4) \quad \mathbf{1}\{u \sim_\sigma v\} = \frac{1}{2}(1 + c(\sigma(uv)) - c(\sigma)), \quad \varepsilon = (R((uv)) - I)c.$$

Step 2: the Fourier coefficients of the cycle count. c is a class function, $c = \sum_{\lambda} \hat{c}_{\lambda} \chi_{\lambda}$ with $\hat{c}_{\lambda} = \langle c, \chi_{\lambda} \rangle = \frac{1}{N!} \sum_{\sigma} c(\sigma) \chi_{\lambda}(\sigma)$. The generating function of c against a character is the content polynomial [9, Ch. I, §1 Ex. 11, §3 Ex. 4 and §7]:

$$\sum_{\sigma \in S_N} x^{c(\sigma)} \chi_{\lambda}(\sigma) = d_{\lambda} \prod_{\square \in \lambda} (x + \text{cont}(\square))$$

(both sides equal $N! s_{\lambda}(1^x)$; for $\lambda = (N)$ it is $x(x+1) \cdots (x+N-1)$). Differentiating at $x = 1$,

$$\hat{c}_{\lambda} = \frac{d_{\lambda}}{N!} \sum_{\square \in \lambda} \prod_{\square' \neq \square} (1 + \text{cont}(\square')).$$

A factor $1 + \text{cont}(\square')$ vanishes exactly when $\text{cont}(\square') = -1$, i.e. for the boxes $(i+1, i)$ on the first subdiagonal. If λ contains two of them, every term of the sum vanishes. If it contains none, $\lambda = (N)$ and $\hat{c}_{(N)} = H_N$, the harmonic number (the mean of c). If it contains exactly one, namely $(2, 1)$ but not $(3, 2)$, then $\lambda_2 \geq 1$ and $\lambda_3 \leq 1$, i.e. $\lambda = (a, b, 1^j)$ with $a \geq b \geq 1$, and only the term $\square = (2, 1)$ survives:

$$\hat{c}_{\lambda} = \frac{d_{\lambda}}{N!} \prod_{\square \neq (2,1)} (1 + \text{cont}(\square)) = \frac{d_{\lambda}}{N!} a! (b-1)! (-1)^j j!,$$

since the contents are $0, \dots, a-1$ in the first row, $0, \dots, b-2$ in the second row without its first box, and $-2, \dots, -(j+1)$ down the column. The hook lengths of $(a, b, 1^j)$ are $a+j+1$ at $(1, 1)$; $a-q+2$ at $(1, q)$ for $2 \leq q \leq b$; $a-q+1$ at $(1, q)$ for $q > b$; $b+j$ at $(2, 1)$; $b-q+1$ at $(2, q)$ for $q \geq 2$; and $j, j-1, \dots, 1$ down the column. (The expansion of c itself, i.e. (5), can also be read off [1, Thm. 3] by summing their expansion of the number of k -cycles over k .) Their product is

$$\frac{N!}{d_{\lambda}} = (a+j+1) \cdot \frac{a!}{(a-b+1)!} \cdot (a-b)! \cdot (b+j) \cdot (b-1)! \cdot j! = \frac{(a+j+1)(b+j)a!(b-1)!j!}{a-b+1},$$

and therefore

$$(5) \quad \hat{c}_{\lambda} = (-1)^j \frac{a-b+1}{(a+j+1)(b+j)} \quad (\lambda = (a, b, 1^j)), \quad \hat{c}_{\lambda} = 0 \text{ for all other } \lambda \neq (N).$$

Step 3: the projections. Π_{λ} is right convolution by a central element of the group algebra, so it commutes with every $R(\rho)$. By (4),

$$\Pi_{\lambda} \varepsilon = (R((uv)) - I) \Pi_{\lambda} c = \hat{c}_{\lambda} (R((uv)) - I) \chi_{\lambda},$$

which vanishes for $\lambda = (N)$ (consistent with $\langle \varepsilon \rangle = 0$, i.e. $\mathbb{P}[u \sim v] = \frac{1}{2}$ under the uniform measure, as $\chi_{(N)} \equiv 1$). Characters are real and satisfy the generalised orthogonality relation $\frac{1}{N!} \sum_{\sigma} \chi_{\lambda}(\sigma) \chi_{\lambda}(\sigma \rho) = \chi_{\lambda}(\rho)/d_{\lambda}$, so

$$\|(R((uv)) - I) \chi_{\lambda}\|^2 = 2 \|\chi_{\lambda}\|^2 - 2 \frac{\chi_{\lambda}((uv))}{d_{\lambda}} = 2(1 - r_{\lambda}).$$

Hence $\nu_{\varepsilon}(\lambda) = 2\hat{c}_{\lambda}^2(1 - r_{\lambda})$, which with (5) is (2).

Step 4: the eigenvalue. In (1) the first row contributes $\sum_{q=1}^a (q-1) = a(a-1)/2$, the second row $\sum_{q=1}^b (q-2) = b(b-3)/2$, and the column $\sum_{i=1}^j (-1-i) = -j(j+3)/2$. $\square$

Remark 2.1. The theorem was checked by brute force over S_N for $N \leq 7$ (applying P to ε directly and comparing $\|P^d \varepsilon\|^2$ and $\langle \varepsilon, P^d \varepsilon \rangle$, $d \leq 4$, with (3); e.g. for $N = 6$, $d = 1$ both give 799/3375 and 17/45), and $\sum_{\lambda} \nu_{\varepsilon}(\lambda) = 1$ was checked to 10^{-12} for $N \leq 4000$.

3. CONSEQUENCES

3.1. The two-point correlation and the exact rate. Write $\Phi(m) := \mathbb{E}[\varepsilon(\sigma_0)\varepsilon(\sigma_m)] = \langle \varepsilon, P^m \varepsilon \rangle$ for the stationary walk, and index the shapes of Theorem 1.1 by the number $k = b + j = N - a$ of boxes outside the first row.

Corollary 3.1. *For all $N \geq 4$ and $m \geq 1$,*

$$0 \leq \|P^d \varepsilon\|^2 = \Phi(2d), \quad |\Phi(m)| \leq \frac{12}{m} + 32 \left(\frac{4 \log(m+1)}{m} + \frac{3}{N} \right)^4 + \frac{1}{(m+1)^4}.$$

Proof. Fix k and b , so $a = N - k$, $j = k - b$, and $N(N-1)(1-r_\lambda) = k(2N-k-1) + j^2 + 3j - b^2 + 3b$.

Level mass. Since $a - b + 1 \leq a + j + 1$, $c_\lambda \leq 1/k$. For $k \leq N - 3$, $j^2 + 3j - b^2 + 3b \leq k^2 + 3k \leq kN$, so $1 - r_\lambda \leq 3k/(N-1) \leq 6k/N$; for $k \geq N - 2$ trivially $1 - r_\lambda \leq 2 \leq 6k/N$. Thus $\nu_\varepsilon(\lambda) \leq 12/(kN)$, and as there are at most k shapes with a given k , the total mass at level k is at most $12/N$.

Spectral gap at level k . Since $b \leq k$ and $b \leq a = N - k$, $b^2 \leq k(N - k)$, so $N(N-1)(1-r_\lambda) \geq k(2N - k - 1) - k(N - k) = k(N - 1)$, i.e. $r_\lambda \leq 1 - k/N$ for every shape. Hence

$$\sum_{\lambda: r_\lambda \geq 0} \nu_\varepsilon(\lambda) r_\lambda^m \leq \frac{12}{N} \sum_{k \geq 1} e^{-km/N} = \frac{12}{N} \frac{1}{e^{m/N} - 1} \leq \frac{12}{m}.$$

The bottom of the spectrum. We may assume $\delta := 4 \log(m+1)/m \leq \frac{1}{2} - \frac{3}{N}$, since otherwise the right-hand side exceeds $2 \geq |\Phi(m)|$. For $r_\lambda < 0$ we use $|r_\lambda|^m \leq (1 - \delta)^m$ unless $1 + r_\lambda \leq \delta$. Now $N(N-1)(1+r_\lambda) = N(N-1) - j(j+3) + a(a-1) + b(b-3)$, and with $j = N - a - b$ one checks $N(N-1) - j(j+3) = (a+b)(N+j) - N - 3j$ and $a(a-1) + b(b-3) \geq -2$; for $a+b \geq 3$ this gives $N(N-1)(1+r_\lambda) \geq (a+b-1)N - 2$ (the only shape with $a+b = 2$ is the sign character (1^N) , with $r = -1$ and $\nu_\varepsilon = 4/(N^2(N-1)^2)$). So $1 + r_\lambda \leq \delta$ forces $a+b \leq \delta N + 3$; such shapes have $\nu_\varepsilon(\lambda) \leq 4c_\lambda^2 \leq 64(\delta N + 3)^2/N^4$ (as $a+j+1$ and $b+j$ are both $\geq N/2$ once $\delta N + 3 \leq N/2$, which holds whenever the bound is non-trivial), and there are at most $(\delta N + 3)^2/2$ of them. Hence $\nu_\varepsilon([-1, -1 + \delta]) \leq 32(\delta + 3/N)^4$, and $(1 - \delta)^m \leq e^{-\delta m} = (m+1)^{-4}$. $\square$

The rate $1/m$ is sharp, and the constant is 2:

Corollary 3.2 (continuum limit). *As $N \rightarrow \infty$, the spectral measure ν_ε converges weakly to the law of $r = 1 - 2(1 - \alpha)(1 - \beta)$ under the probability density*

$$(6) \quad \frac{4(\alpha - \beta)^2}{(1 - \alpha)(1 - \beta)} d\alpha d\beta \quad \text{on } \{\alpha \geq \beta \geq 0, \alpha + \beta \leq 1\},$$

whose mass within δ of $r = 1$ is $2\delta(1 + o(1))$ as $\delta \rightarrow 0$ and within δ of $r = -1$ is $O(\delta^4)$. Consequently

$$(7) \quad \lim_{N \rightarrow \infty} \Phi(m) = \iint \frac{4(\alpha - \beta)^2}{(1 - \alpha)(1 - \beta)} (1 - 2(1 - \alpha)(1 - \beta))^m d\alpha d\beta = \frac{2}{m+1} - \frac{3}{2m^2} + O\left(\frac{1}{m^3}\right).$$

Proof. Put $a = \alpha N$, $b = \beta N$, $j = (1 - \alpha - \beta)N$. Then $c_\lambda = \frac{(\alpha - \beta)N + 1}{((1 - \beta)N + 1)(1 - \alpha)N}$, $1 - r_\lambda = 2(1 - \alpha)(1 - \beta) + O(1/N)$ uniformly, and the shapes in a cell $d\alpha d\beta$ number $N^2 d\alpha d\beta$, so ν_ε is a Riemann sum for (6); the total mass is 1 by Theorem 1.1. Near $r = 1$ put $s = 1 - \alpha$: the region is $\beta \leq s$, the density is $4/s + O(1)$, and $1 - r = 2s(1 - \beta) = 2s + O(s^2)$, so the mass with $1 - r \leq \delta$ is $\int_0^{\delta/2} 4 ds (1 + O(\delta)) = 2\delta + O(\delta^2)$; For the expansion, substitute $t = 1 - \beta = 1 - w$ with $0 \leq w \leq s$: the integrand is $4(1 - s - w)^2(1 - 2s + 2sw)^m / (s(1 - w))$, and expanding in w and s (the mass is carried by $s = O(1/m)$) gives $\int_0^s (\cdots) dw = \frac{4}{s}(1 - 2s)^m (s - \frac{5}{2}s^2 + ms^3 + O(s^3 + m^2 s^5))$; with $\int_0^{1/2} s^k (1 - 2s)^m ds = 2^{-k-1} k! m! / (m + k + 1)!$ this integrates to $\frac{2}{m+1} - \frac{5/2}{(m+1)(m+2)} +$

$\frac{m}{(m+1)(m+2)(m+3)} + O(m^{-3})$. (Numerically $m^2(\lim_N \Phi(m) - 2/(m+1)) \rightarrow -1.500$.) Near $r = -1$ both α and β are $O(\delta)$ and the density is $O(\delta^2)$ on a set of area $O(\delta^2)$. $\square$

Remark 3.3. The mass 2δ at the top of the spectrum has a transparent probabilistic meaning. Under the uniform measure the pair of cycle lengths $(|C_u|, |C_v|)$ is well understood: the smaller of the two relevant masses (the shorter arc between u and v when $u \sim v$, the smaller of the two cycles otherwise) has density 4 at 0^+ on the scale of N (density 2 from each of the two cases, by the conditional laws recalled in the proof of Corollary 3.4), and a pair whose smaller mass is g keeps its status for $\asymp N/g$ steps, since no transposition touches the small part. Thus the slow modes of ε are the “trapped” pairs, and the linear spectral density is the linear density of the gap. Theorem 1.1 shows that this trapping is the *only* slow mechanism and identifies the constant.

3.2. One step.

Corollary 3.4. *Let σ be uniform on S_N and τ an independent uniform transposition. Then*

$$\mathbb{P}[u \sim_\sigma v \text{ and } u \not\sim_{\sigma\tau} v, \text{ or vice versa}] = \frac{(N+1)(N+2)}{6N(N-1)}, \quad \Phi(1) = 1 - \frac{(N+1)(N+2)}{3N(N-1)}.$$

Proof. $\Phi(1) = 1 - 2q$ where q is the probability in question, and $\Phi(1) = \sum_\lambda \nu_\varepsilon(\lambda) r_\lambda$ by (3). The sum is a rational function of N which can be evaluated in closed form, but it is quicker to compute q directly: conditionally on $u \sim v$ with common cycle of length ℓ (probability $(\ell-1)/(N(N-1))$ for each $\ell \geq 2$, the $u \rightarrow v$ distance x being uniform on $\{1, \dots, \ell-1\}$), τ separates the pair iff its two points lie on different arcs, probability $2x(\ell-x)/(N(N-1))$; conditionally on $u \not\sim v$ with cycle lengths (p, q') (probability $\frac{1}{N} \cdot \frac{N-p}{N-1} \cdot \frac{1}{N-p}$ for each admissible pair, by choosing u 's cycle and then v 's), τ merges them with probability $2pq'/(N(N-1))$. Summing the two finite sums gives $q \cdot \binom{N}{2} = \frac{(N+1)(N+2)}{12}$. $\square$

So one random transposition changes the status of a fixed pair with probability $\frac{1}{6} + \frac{2}{3N} + O(N^{-2})$, and $\Phi(1) \rightarrow \frac{2}{3}$.

4. GENERALISATIONS

4.1. Other conjugation-invariant walks. Nothing in Steps 1–3 refers to the transposition walk except the final evaluation $\chi_\lambda((uv))/d_\lambda = r_\lambda$, which comes from the *definition* of ε , not from the walk. Hence for any random walk $\sigma_{t+1} = \sigma_t \rho_{t+1}$ with ρ drawn from a conjugation-invariant law μ on S_N (random k -cycles, random involutions, the $(N-k)$ -cycle walk, ...), whose eigenvalues are $r_\lambda^\mu = \sum_\rho \mu(\rho) \chi_\lambda(\rho)/d_\lambda$,

$$\mathbb{E}[\varepsilon(\sigma_0) \varepsilon(\sigma_m)] = \sum_{\lambda=(a,b,1^j)} 2c_\lambda^2 (1 - r_\lambda) (r_\lambda^\mu)^m,$$

with c_λ and r_λ as in Theorem 1.1. The spectral measure is intrinsic to the function ε ; only the moments change.

4.2. Non-stationary starts. For a fixed start σ_0 , $\Pi_\lambda \varepsilon = \hat{c}_\lambda (R((uv)) - I) \chi_\lambda$ gives

$$(8) \quad \mathbb{E}[\varepsilon(\sigma_0 \tau_1 \cdots \tau_m)] = (P^m \varepsilon)(\sigma_0) = \sum_{\lambda=(a,b,1^j)} r_\lambda^m \hat{c}_\lambda [\chi_\lambda(\sigma_0(uv)) - \chi_\lambda(\sigma_0)],$$

with $\hat{c}_\lambda$ carrying the sign $(-1)^j$ of (5). At $m = 0$ this reads $\sum_\lambda \hat{c}_\lambda [\chi_\lambda(\sigma_0(uv)) - \chi_\lambda(\sigma_0)] = \varepsilon(\sigma_0)$, which for $\sigma_0 = \text{id}$ (where $\chi_\lambda(\text{id} \cdot (uv)) - \chi_\lambda(\text{id}) = -d_\lambda(1 - r_\lambda)$ and $\varepsilon(\text{id}) = -1$) is the identity $\sum_{\lambda=(a,b,1^j)} \hat{c}_\lambda d_\lambda (1 - r_\lambda) = 1$. When σ_0 is an N -cycle, $\chi_\lambda(\sigma_0)$ vanishes unless λ is a hook, and

$\sigma_0(uv)$ has cycle type $(\ell, N - \ell)$, where the characters of the shapes $(a, b, 1^j)$ are given by the Murnaghan–Nakayama rule in terms of two rim hooks; (8) then becomes an explicit finite sum for the probability that two points at distance ℓ on an N -cycle are still (or again) in a common cycle after m random transpositions.

4.3. The structure behind the formula. The identity (4) says that the same-cycle indicator is a first difference of the cycle count, and the cycle count is supported on the shapes $(a, b, 1^j)$ because its generating function is the content polynomial, whose derivative at $x = 1$ survives only one subdiagonal box. More generally, for any $g \in S_N$ the function $\sigma \mapsto c(\sigma g) - c(\sigma)$ has spectral measure $\hat{c}_\lambda^2 \|(R(g) - I)\chi_\lambda\|^2 = 2\hat{c}_\lambda^2(1 - \chi_\lambda(g)/d_\lambda)$, supported on the same shapes. Functions such as $\mathbf{1}\{u_1 \sim u_2 \sim u_3\}$ are not of this form, and we do not know whether their spectral measures are equally explicit.

Remark 4.1 (the Latin square problem). Let L be a Latin square of order n , let j, j' be two columns, and let π be the permutation of the rows sending r to the row r' with $L(r', j') = L(r, j)$ (the “column cycles” of the pair). Whether two given rows lie in a common cycle of π is a bit that appears in the switching identities of Cavenagh, Greenhill and Wanless for the distribution of the second row of a random Latin square [6], and the re-randomisation method of Kwan, Petrova and Sawhney [8] randomises it by switching a pool of disjoint intercalates through the columns j, j' , each of which multiplies π by a transposition of rows. If the base permutation is uniform and the pool consists of k uniform disjoint transpositions, the bit’s conditional bias given the pool, $\text{bias} = \mathbb{P}_S[\text{bit}] - \frac{1}{2}$ over uniform subsets S of the pool, satisfies $4\mathbb{E}[\text{bias}^2] = \mathbb{E}_M \Phi(M) + O(k^2/n)$ with $M \sim \text{Bin}(k, \frac{1}{2})$ (sketch: with S, T independent uniform subsets, $\varepsilon_S \varepsilon_T = \varepsilon(\alpha) \varepsilon(\alpha \tau_{S \Delta T})$ for the uniform $\alpha = \sigma \tau_S$, and $|S \Delta T|$ disjoint uniform transpositions avoiding u, v are coupled to iid ones at cost $O(k^2/n)$), so that $\mathbb{E}|\text{bias}| = O(k^{-1/2})$ by Corollary 3.1; this is the estimate that the present computation was made for.

REFERENCES

- [1] G. Alon and G. Kozma, The probability of long cycles in interchange processes, *Duke Math. J.* 162 (2013), 1567–1585.
- [2] D. Arcona, Representation theory and cycle statistics for random walks on the symmetric group, arXiv:2512.13969 (2026).
- [3] N. Berestycki and R. Durrett, A phase transition in the random transposition random walk, *Probab. Theory Related Fields* 136 (2006), 203–233.
- [4] N. Berestycki and G. Kozma, Cycle structure of the interchange process and representation theory, *Bull. Soc. Math. France* 143 (2015), 265–280.
- [5] N. Berestycki, O. Schramm and O. Zeitouni, Mixing times for random k -cycles and coalescence-fragmentation chains, *Ann. Probab.* 39 (2011), 1815–1843.
- [6] N. J. Cavenagh, C. Greenhill and I. M. Wanless, The cycle structure of two rows in a random Latin square, *Random Structures Algorithms* 33 (2008), 286–309.
- [7] P. Diaconis and M. Shahshahani, Generating a random permutation with random transpositions, *Z. Wahrsch. Verw. Gebiete* 57 (1981), 159–179.
- [8] M. Kwan, A. Petrova and M. Sawhney, Parities in random Latin squares, arXiv:2509.13125 (2025).
- [9] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, 2nd ed., Oxford University Press, 1995.
- [10] O. Schramm, Compositions of random transpositions, *Israel J. Math.* 147 (2005), 221–243.

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